

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		iron(III) oxide		1		1		
		(ii)		award (1) for either of following this happens as a lower temperature than the melting point of aluminium oxide the melting point of cryolite is lower than the melting point of aluminium oxide accept lowers the melting point of aluminium oxide accept increases conductivity of the electrolyte neutral answer – lowers energy cost	1			1		
		(iii)		award (1) for either of following anode / graphite reacts with oxygen / air anode / graphite is burnt away / used up award (1) for either of following forms carbon dioxide $C + O_2 \rightarrow CO_2$	2			2		
		(iv)		$2Al_2O_3 \longrightarrow \boxed{4} Al + 3 O_2$ balancing aluminium atoms (1) product (1) discrete marks		2		2		